Machine Learning Assignment
Assignment 7: Diabetes Prediction using Decision Tree
SDG 3 – Good Health and Well-being

1. Problem Statement

Design a Machine Learning model to predict whether a patient is likely to have diabetes based on medical attributes. Early prediction helps doctors provide timely treatment and reduces complications.

2. Relevant SDG

SDG 3 – Good Health and Well-being aims to ensure healthy lives and promote well-being. Diabetes prediction supports preventive healthcare.

3. Background Study

Diabetes is a chronic disease affecting millions worldwide. Machine Learning analyzes historical medical records to identify patterns associated with diabetes. Common algorithms include Decision Trees, Logistic Regression, Random Forest, and SVM.

4. Learning Problem

Input Features: Pregnancies, Glucose, Blood Pressure, Skin Thickness, Insulin, BMI, Diabetes Pedigree Function, Age.
Target Output: Diabetes (Positive/Negative).
Data Representation: Tabular dataset using Pandas DataFrame.

5. Dataset

Pima Indians Diabetes Dataset (UCI/Kaggle). Preprocessing includes handling missing values, checking duplicates, feature scaling if required, and train-test split.

6. ML Approach

Decision Tree Learning is selected because it is easy to interpret, handles numerical features well, and matches the Unit I syllabus.

7. Workflow

Collect Dataset → Preprocess Data → Feature Selection → Train Decision Tree → Test Model → Evaluate Accuracy → Predict Diabetes.

8. Algorithm

Load dataset using Pandas; preprocess data; split into training/testing sets; train DecisionTreeClassifier; predict test samples; evaluate using Accuracy, Precision, Recall, F1-score, and Confusion Matrix.

9. NumPy/Pandas Analysis

Use Pandas to inspect the dataset (head, info, describe, null values). Use NumPy for numerical operations such as averages, standard deviation, and array manipulation.

10. Visualization

Create histograms of glucose and BMI, correlation heatmap, diabetes class distribution bar chart, and confusion matrix. Interpret trends to identify influential features.

11. Expected Outcomes

The model predicts diabetes risk with good accuracy, assisting healthcare professionals in early diagnosis and improving patient outcomes.

12. Advantages

Fast prediction; interpretable model; supports preventive healthcare; low computational cost.

13. Limitations

Prediction quality depends on dataset quality; may not generalize to all populations; clinical validation is required.

14. Practical Applications

Hospital decision support, health screening camps, telemedicine, and mobile health applications.

15. Recommendations

Use larger datasets, compare multiple algorithms, apply cross-validation, and update the model periodically.

16. Conclusion

Decision Tree-based diabetes prediction is an effective Machine Learning solution that supports SDG 3 by enabling early diagnosis and better healthcare decisions.

17. Sample Python Libraries

```
import pandas as pd
import numpy as np
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
```

18. References

1. UCI Machine Learning Repository – Pima Indians Diabetes Dataset.
2. Scikit-learn Documentation.
3. Pandas Documentation.
4. NumPy Documentation.